Student Name: Aarav Sharma | Roll Number: ROLL-101

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Ans 1 Newton's Second Law: The rate of change of linear momentum of an object is directly proportional to the unbalanced external force acting on it.

Derivation: P = mv

F proportional to dp/dt = d(mv)/dt = m(dv/dt) = m*a

Hence, F = m * a.

Ans 2 Given: m = 1500 kg, u = 0, v = 20 m/s, t = 10 s.

(a) Using first equation of motion:

v = u + at => 20 = 0 + a(10) => a = 2 m/s^2.

(b) Net force F = m * a = 1500 kg * 2 m/s^2 = 3000 N.

Ans 3

Work: Work is said to be done by a force when it causes displacement.

W = Force x Displacement in direction of force = F * d.

Given: F = 50 N, d = 6 m.

W = 50 * 6 = 300 Joules (J).

Ans 4

1. Kinetic Energy: The energy possessed by a body due to its motion.

Example: A fast moving cricket ball or flowing water.

2. Potential Energy: The energy stored due to relative position or shape.

Example: Water stored at height in a dam reservoir.